

University of Karachi

Department of Computer Science — UBIT

SMART MEAL PLANNER & GROCERY AUTOMATOR

Project Documentation

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Course: **Internet Application Development**

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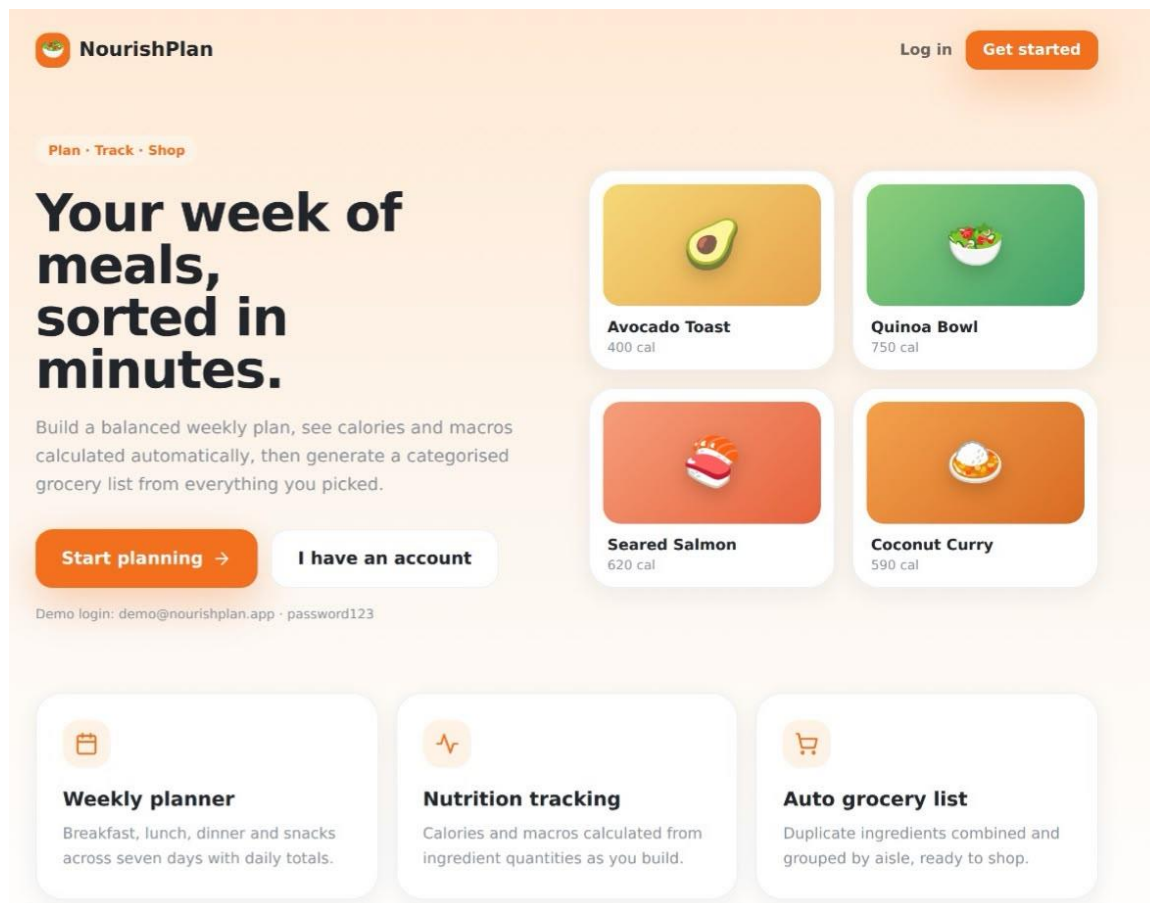
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1. Project Overview

Smart Meal Planner & Grocery Automator is a full-stack Software-as-a-Service (SaaS) web application designed to simplify meal planning, nutrition tracking, and grocery shopping. The platform allows users to create a personal dietary profile, set nutrition goals, add or select meals, track calories and macronutrients, and automatically generate a grocery list from a weekly meal plan.

The application combines three common activities into one platform: planning meals, understanding food nutrition, and preparing a shopping list. This makes the system useful for students, working professionals, fitness-focused users, and anyone who wants to organize their weekly meals.

Item	Details
Project Name	Smart Meal Planner & Grocery Automator
Document Type	Software Project Documentation
Application Type	Full-Stack SaaS Web Application
Frontend	Next.js, React, TypeScript, Tailwind CSS
Backend	NestJS, TypeScript, REST API
Database	MongoDB with Mongoose
Primary Users	Individuals interested in meal planning and nutrition tracking
Version	1.0
Status	Proposed / MVP Specification



2. Problem Statement

Many people struggle to plan daily meals, track nutrition, and maintain healthy eating routines. Manual meal planning, scattered nutrition information, and separate grocery lists can lead to wasted time, unnecessary purchases, and difficulty achieving health goals.

The proposed system addresses:

- Difficulty creating balanced weekly meal plans.
- Lack of simple calorie and nutrient tracking.
- Manual grocery list preparation.
- Difficulty tracking meals and remaining daily calories.
- Limited personalization for dietary preferences, allergies, and fitness goals.

3. Project Objectives

- Develop a responsive web-based SaaS application using Next.js, NestJS, and MongoDB.
- Allow users to register, log in, and manage their dietary profiles.
- Provide meal creation and meal management functionality.
- Automatically calculate nutrition values from ingredient quantities.
- Provide a weekly meal planner with daily calorie and macro summaries.
- Generate an organized grocery list from the selected weekly meals.

- Support free and premium subscription concepts.
- Create a scalable architecture that can later support AI meal generation and grocery delivery integrations.

4. Proposed Solution

The proposed solution is a centralized meal planning platform. After registration, the user enters basic information such as age, height, weight, activity level, dietary preference, allergies, and fitness goal. The system estimates a daily calorie target and allows the user to create a weekly meal plan.

Users can add meals manually by entering ingredients and quantities. The system retrieves nutrition values from its food database and calculates the total calories and macronutrients for each meal. When the user completes the weekly plan, the grocery module combines ingredients from all selected meals and creates a categorized checklist.

5. Scope of the Project

5.1 In-Scope Features

- User registration and login.
- JWT-based authentication.
- User dietary profile.
- Meal creation, editing, viewing, and deletion.
- Weekly meal planning.
- Daily nutrition dashboard.

5.2 Out-of-Scope for the Initial MVP

- Real payment processing.
- Direct grocery delivery checkout.
- Medical or clinical nutrition advice.
- Barcode scanning and image-based food recognition.

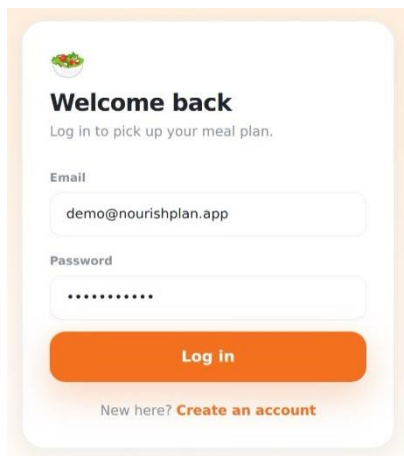
These features may be added in future versions after the core MVP is stable.

6. Functional Requirements

6.1 Authentication and Account Management

- The system shall allow a new user to register with name, email, and password.
- The system shall validate email format and password requirements.
- Passwords shall be hashed before storage.

- The system shall allow users to log in and log out.



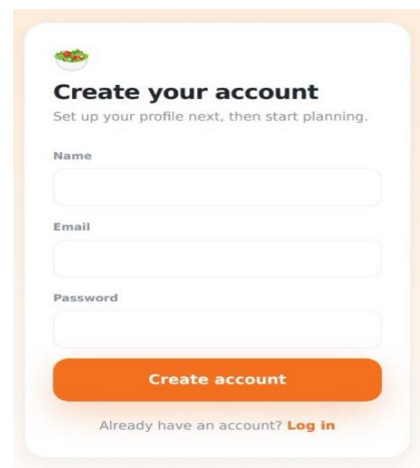
Welcome back
Log in to pick up your meal plan.

Email
demo@nourishplan.app

Password
.....

Log in

New here? [Create an account](#)



Create your account
Set up your profile next, then start planning.

Name
[Input field]

Email
[Input field]

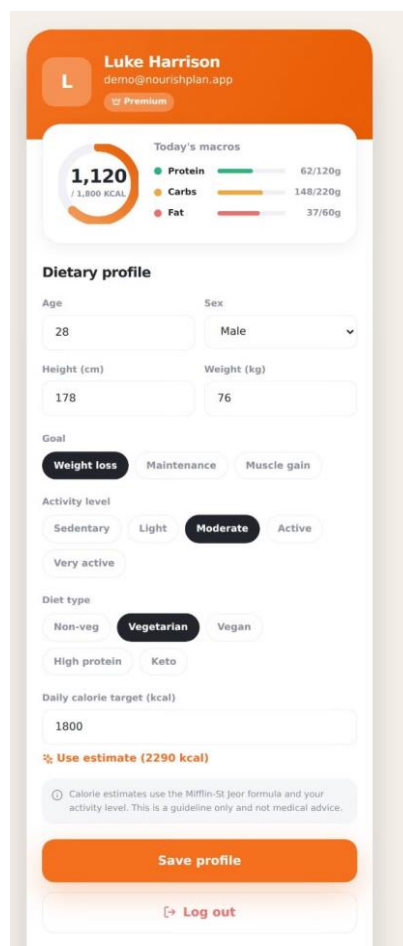
Password
[Input field]

Create account

Already have an account? [Log in](#)

6.2 Dietary Profile

- Users shall be able to enter age, height, weight, activity level, and fitness goal.
- Users shall be able to select a diet type such as vegetarian, non-vegetarian, or high-protein.
- Users shall be able to add allergies and disliked ingredients.



Luke Harrison
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[Premium](#)

Today's macros

1,120 / 1,800 KCAL

- Protein: 62/120g
- Carbs: 148/220g
- Fat: 37/60g

Dietary profile

Age: 28 Sex: Male

Height (cm): 178 Weight (kg): 76

Goal: **Weight loss** Maintenance Muscle gain

Activity level: Sedentary Light **Moderate** Active

Diet type: Non-veg **Vegetarian** Vegan

High protein Keto

Daily calorie target (kcal): 1800

[Use estimate \(2290 kcal\)](#)

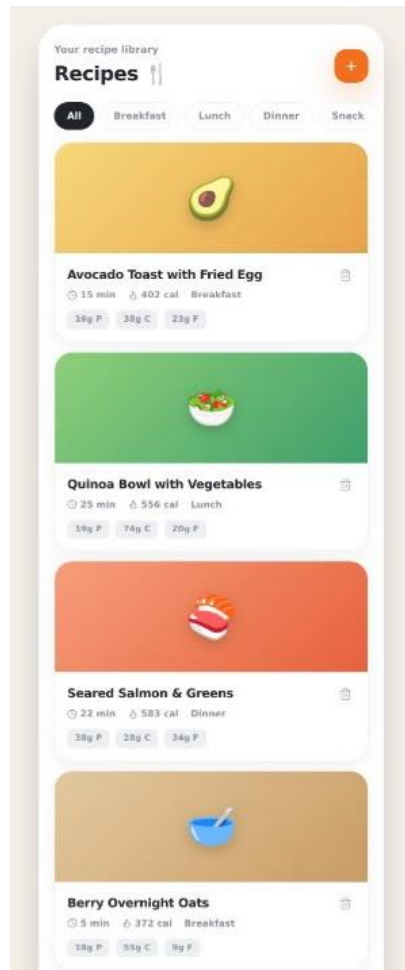
Calorie estimates use the Mifflin-St Jeor formula and your activity level. This is a guideline only and not medical advice.

Save profile

[Log out](#)

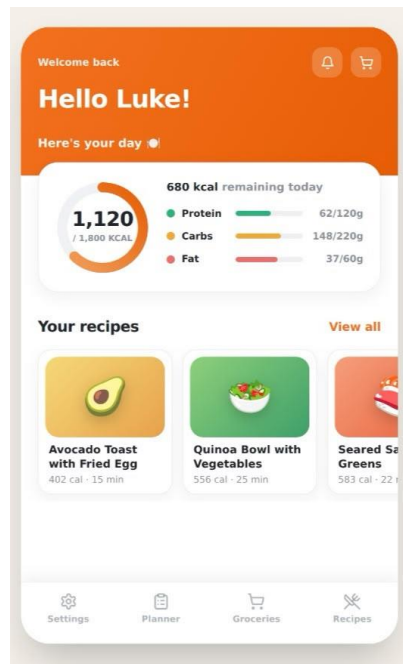
6.3 Meal Management

- Users shall be able to create a meal.
- Users shall be able to select a meal type: breakfast, lunch, dinner, or snack.
- Users shall be able to add multiple ingredients to a meal.



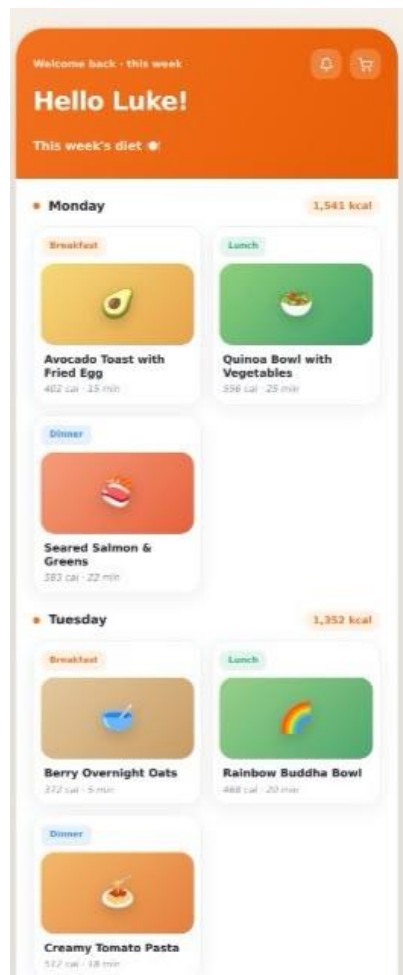
6.4 Nutrition Tracking

- The system shall display calories per meal.
- The system shall display protein, carbohydrates, fats, and fiber where available.
- Users shall be able to log meals as consumed.
- The dashboard shall show daily calories consumed and remaining.



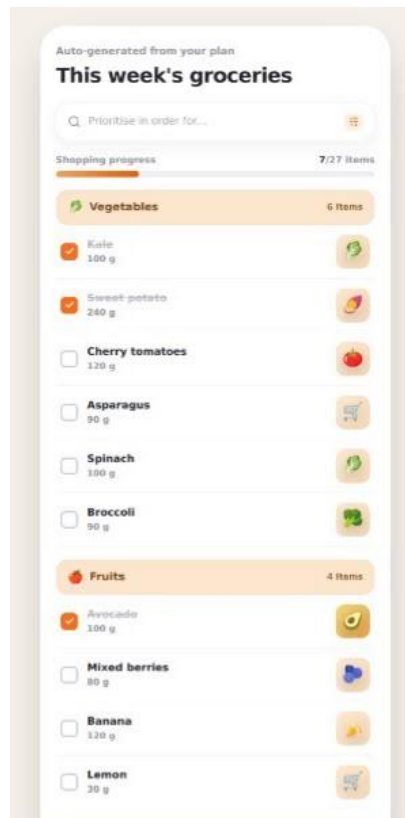
6.5 Weekly Meal Planner

- Users shall be able to create a seven-day meal plan.
- Each day shall support breakfast, lunch, dinner, and snacks.
- Users shall be able to add, replace, or remove meals.



6.6 Grocery List Automation

- The system shall extract ingredients from the selected meal plan.
- Duplicate ingredients shall be combined.
- The system shall normalize compatible units where possible.
- Grocery items shall be grouped into categories such as vegetables, fruits, grains, dairy, meat, and pantry.



6.7 Subscription Features

- The system shall support free and premium plan types.
- Free users shall have access to basic meal planning and grocery list features.
- Premium users shall have access to advanced features such as unlimited recipes, advanced analytics, and future integrations.
- Payment processing may be mocked in the MVP and integrated later.

7. User Roles

Role	Responsibilities
Guest	View landing page, features, pricing, and registration/login pages.
Registered User	Manage profile, meals, meal plans, nutrition logs, and grocery lists.

Premium User	Access premium features according to subscription status.
Administrator (Optional)	Manage food items, monitor users, and manage subscription settings.

8. Technology Stack

Technology	Purpose
Next.js	Frontend framework and routing.
React	Reusable UI components.
TypeScript	Type-safe development.
Tailwind CSS	Responsive and consistent styling.
NestJS	Backend framework and modular REST API.
MongoDB	NoSQL database.
Mongoose	MongoDB object modeling and schema management.
JWT	Authentication and protected routes.
bcrypt	Password hashing.
Recharts	Nutrition and progress charts.
Postman	API testing.
Git/GitHub	Version control and collaboration.

9. Frontend Design

9.1 Main Pages

Route	Purpose
/	Landing page and project introduction.
/login	User login.
/register	Account registration.
/dashboard	Calories, macros, recent meals, and quick actions.
/profile	Dietary profile and goals.
/meals	List and manage meals.
/meals/create	Create a custom meal.

/meals/[id]	View meal details and nutrition.
/planner	Weekly meal planning calendar.
/grocery	Automated grocery list and checklist.
/history	Previous plans and nutrition history.
/pricing	Free and premium plans.

9.2 Suggested Component Structure

The frontend should be organized into reusable components to make the code maintainable and easy to extend.

```

components/ ├── Navbar.tsx ├── Sidebar.tsx ├── MealCard.tsx ├──
MealForm.tsx ├── IngredientInput.tsx ├── NutritionCard.tsx ├──
MacroProgress.tsx ├── CalorieChart.tsx ├── WeeklyPlanner.tsx ├──
GroceryList.tsx ├── GroceryItem.tsx ├── ProtectedRoute.tsx ├──
LoadingSpinner.tsx

```

10. Backend Design

10.1 NestJS Modules

Module	Responsibilities
AuthModule	Registration, login, JWT generation, and authentication guards.
UsersModule	User profile and dietary preferences.
MealsModule	Meal CRUD and ingredient assignment.
IngredientsModule	Food items and nutrition data.
NutritionModule	Nutrition calculations, logs, and summaries.
MealPlansModule	Weekly plan creation and management.
GroceryModule	Ingredient aggregation and grocery checklist.
SubscriptionsModule	Free/premium plan access control.

11. Core Business Logic

11.1 Nutrition Calculation

Each ingredient should store nutrition values per 100 grams or per standard serving. When a user enters a quantity, the backend scales the nutrition values and adds the results for all ingredients.

$$\text{Nutrient Total} = \Sigma (\text{Ingredient Quantity} \div 100 \times \text{Nutrient per 100g})$$

For example, if an ingredient contains 165 calories per 100g and the user adds 150g, the calculated calories are $150 \div 100 \times 165 = 247.5$ calories. The same process is applied to protein, carbohydrates, fats, and fiber.

11.2 Estimated Calorie Target

The application may use a standard BMR and activity-based calculation to estimate a user's daily calorie target. The result is an estimate and should be editable by the user.

$$\begin{aligned} \text{BMR (Male)} &= 10W + 6.25H - 5A + 5 \\ \text{BMR (Female)} &= 10W + 6.25H - 5A \\ - 161 \text{ Estimated Daily Energy} &= \text{BMR} \times \text{Activity Factor} \end{aligned}$$

W represents weight in kilograms, H represents height in centimeters, and A represents age in years. The system should avoid presenting the result as medical advice or enforcing unsafe calorie limits.

12. User Workflow

1. User opens the application.
2. User creates an account or logs in.
3. User completes the dietary profile.
4. System estimates a daily calorie target.
5. User adds meals or selects existing meals.
6. User creates a seven-day meal plan.
7. System calculates daily calories and macros.
8. User logs consumed meals throughout the day.
9. User clicks Generate Grocery List.
10. System combines ingredients and creates a categorized checklist.
11. User marks purchased items or exports the list.

13. SaaS Monetization

The application may use a monthly or yearly subscription for premium features. During the university MVP, subscription access can be simulated using a plan field in the user document.

Feature	Free Plan	Premium Plan
Meal planning	Limited	Unlimited
Custom meals	Limited	Unlimited
Calorie counter	Basic	Advanced
Grocery list	Available	Advanced export/integration
Nutrition history	Short history	Full history
AI meal generation	Not available	Future premium feature
Grocery delivery integration	Not available	Future premium feature

14. Non-Functional Requirements

Requirement	Description
Performance	Pages and APIs should respond quickly under normal usage.
Security	Passwords must be hashed and protected routes must require authentication.
Scalability	The modular backend should support future features.
Usability	The interface should be simple and easy to understand.
Responsiveness	The application should work on desktop, tablet, and mobile.
Maintainability	Code should use reusable components and clear modules.
Reliability	Invalid quantities and missing nutrition values should be handled safely.
Privacy	User information should be stored securely and deletion should be supported.

15. Expected Outcomes

After completion, the project is expected to deliver a functional SaaS web application where users can manage meals, understand their nutrition intake, create weekly plans, and generate grocery lists. The project will demonstrate practical knowledge of frontend development, backend API design, database modeling, authentication, business logic, and SaaS concepts.

- A working responsive frontend.
- A modular NestJS backend.
- A MongoDB database with related collections.
- Secure user authentication.
- Automatic nutrition calculations.
- A usable weekly meal planner.
- An automated grocery checklist.
- A clear foundation for premium and AI-powered features.

16. Conclusion

Smart Meal Planner & Grocery Automator is a practical and scalable full-stack SaaS project. It solves a real-world problem by connecting meal planning, calorie tracking, nutrition analysis, and grocery list automation in one application. The proposed architecture using Next.js, NestJS, and MongoDB is suitable for rapid development and future expansion. The MVP should focus on the most important user journey: create a profile, add meals, calculate nutrition, build a weekly plan, and generate a grocery list.

Appendix A: Sample Meal Nutrition

Ingredient	Quantity	Calories	Protein	Carbs	Fat
Chicken breast	150g	248	46g	0g	5.4g
Cooked rice	150g	195	4g	42g	0.5g
Broccoli	100g	35	2.4g	7g	0.4g
Olive oil	1 tsp	40	0g	0g	4.5g
Total		518	52.4g	49g	10.8g

Note: Sample nutrition values are illustrative. Production values should be obtained from a reliable nutrition database and should account for preparation method and serving size.